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FL-IIDS: A novel federated learning-based incremental intrusion detection system

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ABSTRACT

With the advantage of analyzing data of multiple work sites comprehensively while ensuring data privacy, federated learning-based intrusion detection systems (IDS) are emerging as a distributed intrusion detection paradigm. Most of these IDS are assumed to work on static data. However, in the actual network environment, the practice of setting data as static will lead to the phenomenon known as catastrophic forgetting, where old classes that have already appeared would be forgotten. In this paper, we propose a novel IDS framework called FL-IIDS to effectively address the catastrophic forgetting problem. Firstly, a new loss function is synthetically designed for local model training. With the new function, the class gradient balance loss function assigns different learning weights to data of the new and old classes so that the learning rate of the new classes would decrease and the memory of the overall old classes would be deepened. Moreover, the sample label smoothing loss function leverages the knowledge distillation method to enhance the local model memory for every specific class of old classes. Secondly, the relay client fusing sample reconstruction is employed to mitigate the spread of catastrophic forgetting globally without compromising data privacy. Extensive experimental results on the UNSW-NB15 dataset and the CICIDS2018 dataset show that our proposed framework improves the memory capability for old classes substantially without affecting the detection effectiveness of the IDS for new classes.

1. Introduction

Due to the continuous development of communication technologies and the Internet, data interaction has been required more strongly, especially in areas of healthcare, social media, and the smart industry. And then the amount of network traffic data largely increases [1]. Moreover, these data appear diverse and heterogeneous. It indicates more network vulnerabilities may be exposed to attackers. As a result, the security of the huge traffic data has become an important challenge. Intrusion Detection Systems (IDS), as an important technology for network security, seems a candidate solution, which monitors online traffic in real-time and defends against attacks hidden in network traffic [2]. At the same time, thanks to the development of artificial intelligence technology, deep learning-related methods can improve the ability of IDSs to identify attacks on time, thus protecting devices from intrusions in real time and efficiently. In recent years, the centralized IDS based on deep learning is widely studied and has achieved more excellent research results [3–5]. However, centralized IDS has exhibited its limitations when smart Internet of Things (IoT) applications and edge computing applications are widely deployed. Firstly, the centralized IDS would inevitably deviate from the privacy of edge devices [6], since it collects a large amount of traffic data from sub-users or sub-devices. Secondly, centralized intrusion detection usually consumes significant computational resources and then raises time overhead [7].

Federated learning, proposed by Google in 2016 [8], has been employed by IDS to overcome the drawbacks of centralized IDS. It can coordinate several different sub-devices to jointly complete the construction of global deep learning models and solve the problem of data isolated island without exposing the private data of each sub-device [9]. An IDS that incorporates federated learning can enhance its ability to monitor multiple data sites without compromising the data in each data site. Therefore, federated learning, as a distributed deep learning

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paradigm, is well suited for IDS for the scenarios of IoT and edge computing, and has become a widely used technology [10–15].

However, most of the current research related to federated learning-based intrusion detection is oriented towards static traffic data. They assume that the training data faced by each local client is stationary and non-time-varying. But in the real-world environment, the traffic data available for the local client changes over time. And setting the training data to be static makes it extremely difficult for joint learning models to adapt to this real dynamic environment [16]. In this case, the previous knowledge of the local client would be overwritten by new knowledge constantly, which would significantly compromise the classification performance of the local model for the previous knowledge, a phenomenon known as catastrophic forgetting [17]. Further, catastrophic forgetting can spread from the local model to the global model and then eventually make the global model forget the learned knowledge after many iterations of federated learning.

To address catastrophic forgetting, two types of solutions have been proposed. The first type is to continuously store several samples of each old class that have been learned in the local client so that the old samples could participate in the current model to train together with the new samples. However, it conflicts with the lightweight requirement for resource-constrained devices since it would result in a storage burden for local clients [7,18,19]. The second type is the incremental IDS [20–22]. To our best knowledge, incremental solutions are applicable for centralized intrusion detection and there is no research for federated intrusion detection scenarios.

We propose an innovative Federated Learning-based Incremental Intrusion Detection System (FL-IIDS), which solves the catastrophic forgetting in federated learning. The proposed FL-IIDS focuses on the most common class incremental problems in IDS where data follows the independent identically distributed (i.i.d) among local clients. A recall-based approach is combined with a regularization-based approach to weaken the consequence of local-side catastrophic forgetting. At the same time, a relay client is employed to mitigate the global-side catastrophic forgetting. Our contributions are summarized as follows.

- A dynamic example memory is adapted to mitigate catastrophic forgetting, which enables the local client to identify old classes that have appeared in the training history. It simulates the distribution of traffic data in a real environment and overcomes the concept drift phenomenon to a certain extent.
- A class gradient balance loss and a sample label smoothing loss are constructed to remember both the overall old classes and each specific old class, respectively. The former uses a reweighting approach to adjust the learning rate of the local model for new versus old classes, while the latter uses the best old model stored in the local client to fuse each old class knowledge into the current training labels.
- The relay clients combined with sample reconstruction are designed to select the best old model at the global level so that the proliferation of global catastrophic forgetting is prevented.

The rest of the paper is structured as follows: Section 2 reviews related works. Section 3 gives the structure and the operation of FL-IIDS and exhibits the incremental learning algorithm based on the specific reason of catastrophic forgetting. And then Section 4 shows the corresponding experimental results and analysis. At last, Section 5 discusses the conclusions and gives an outlook on the direction of future work.

2. Related works

2.1. Federated learning-based IDS

In recent years, federated learning-based IDSs have become a research hotspot. In the Ref. [23], a decentralized federated learning approach with an ensembler is proposed and then used for intrusion detection in IoT networks. It utilizes gated recursive units as local models and designs an ensembler based on random forest decision trees. The ensembler optimizes the model parameters of the global model by collecting updates from multiple sources to improve the classification capability of the global model for traffic data. Liu et al. [24] proposed an intrusion detection system for protecting vehicular networks by fusing federated learning techniques with blockchain technology. The system designs a secret-sharing-based mask noise model uploading algorithm based on federated learning to ensure secure uploading of local model parameters and improve model accuracy using a model quality-based evaluation algorithm. In the literature [25], to improve the security of advanced metering infrastructure in smart grid, a hierarchical federated learning-based intrusion detection system is designed to enhance the classification performance of federated learning by using Transformer as an intrusion detection model while ensuring the privacy of smart meter data is not compromised. To address the security problem in large-scale and heterogeneous industrial information-physical systems, a federated deep learning scheme called DeepFed was proposed in the literature [10]. The scheme first constructs an intrusion detection classification model using convolutional neural network and gated recurrent unit, and then indirectly acquires data resources from different working sites using a federated learning framework to build a robust intrusion detection model, and the excellent performance on real datasets from natural gas pipeline systems proves the effectiveness of the scheme. Although they have improved the recognition accuracy with federated learning, the above studies have not mentioned the privacy protection.

To overcome the limitation of the above studies, some proposals with privacy protection have been exhibited. The literature [26] provides a comprehensive evaluation of several existing differential privacy approaches through additive noise techniques based on Gaussian and Laplace distributions, aiming to investigate which DP schemes can better protect the federated learning framework. And this literature uses the dataset ToN-IoT to compare the effect of differential privacy protection under different federal learning frameworks (e.g. FedAvg, Fed+), which contributes valuable experience for the subsequent federal learning privacy protection. Besides, some studies focus on attacks that are more harmful to the federated learning framework and investigate protection measures against such attacks. Zhang et al. [27] proposed a federated learning framework that is immune to poisoning attacks. The online unsupervised poison detection method of this framework can effectively screen poisoned models and reject participation in global aggregation sessions so that the entire federated learning framework is not compromised.

2.2. Incremental learning approach to IDS

The incremental IDS is the focus of the IDS. A dynamic integration incremental learning intrusion detection method is proposed in the literature [28]. It uses relevance vector machines (RVM) as components of the integration module and prunes the RVM components of the integration module in a dynamic data environment using performance metrics as reference thresholds so that the intrusion detection model always maintains the memory of old knowledge. The method achieved dazzling performance on CICIDS2017 as well as the KDD CUP99 dataset. The literature [29] proposes a new framework for incremental intrusion detection systems, which can achieve learning without prior knowledge using incremental clustering algorithms, and which also incorporates transfer learning-related methods so that at each iteration of the model, transfer learning provides incremental knowledge to the target environment, thus accomplishing incremental learning. The literature [21] proposed an incremental learning model called T-DFNN, which consists of multiple deep feedforward neural network models connected in a tree-like structure. The F1 macro mean of this model at CICIDS2017 exceeded 0.85 in each training phase. The literature [30] evaluates the advantages and disadvantages of the two incremental learning methods, Elastic Weight Consolidation (EWC) and Gradient Episode Memory

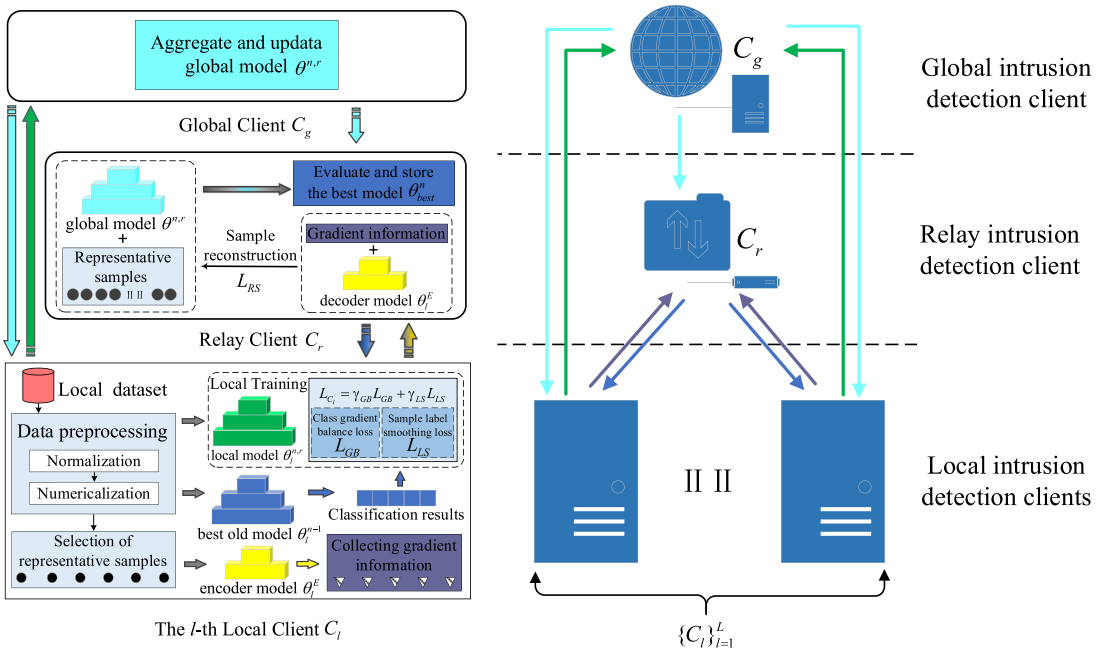


Fig. 1. The complete architecture of FL-IIDS.

(GEM), under the condition that the data are class unbalanced, from the perspective of both class incremental tasks and domain incremental tasks. In the literature [31], an adaptive incremental intrusion detection strategy based on extreme learning machines (ELMs) is proposed. During the learning process, the model can adaptively add hidden neurons to find the best model that fits both new and old knowledge in a dynamic data stream, and experiments demonstrate that the strategy achieves a better performance with a smaller time cost.

In summary, in the research field of federated learning-based intrusion detection, most of the work aims to research robust and secure intrusion detection systems with excellent recognition ability for malicious traffic. Actually, a continuous traffic monitoring mechanism is more meaningful in practical scenes, and the current researches in federated learning-based intrusion detection lack consideration about this. In the research field of incremental intrusion detection, the vast majority of researchers pay more attention to centralized scenes. In distributed and computationally limited application scenarios, these incremental strategies need to ceaselessly transform the model structure according to the different tasks. Thus, there is a need for an incremental intrusion detection model with the adaptation ability for distributed and variable tasks. Our work combines federated learning with incremental learning and designs an incremental learning mechanism that adapts to diverse intrusion detection environments without performing model transformations.

3. Proposed method

FL-IIDS is proposed to solve the catastrophic forgetting faced by federated IDS. It is important to note that the traffic data between each local client in FL-IIDS follows the independent and identically distributed(i.i.d). And we propose a framework that is oriented towards federated aggregation algorithms: FedAvg. In Section 3.1, the overall framework of FL-IIDS and the specific operational flow is introduced. And then the storage mechanism and training algorithm are designed based on the analysis for the reason of the catastrophic forgetting of local models in Section 3.2. At last, in Section 3.3, the solutions for global catastrophic forgetting are proposed.

3.1. Architecture of FL-IIDS

The complete architecture of our proposed FL-IIDS is shown in Fig. 1. The clients are divided into two categories for incremental learning tasks. The first ones are the local clients, which implement a new memory mechanism and an incremental training method. They aim to alleviate the local-side catastrophic forgetting problem. The second ones are the relay clients, which implement the sample reconstruction mechanism to find the best old model They would mitigate the global-side catastrophic forgetting. The left half of Fig. 1 shows the specific process of the lth local client C_l participating in global aggregation to engage in federated learning, and the right half shows the details of the communication between the clients during one global round.

For FL-IIDS, the local client is assumed to receive a series of task flows $T_l = \{T_l^n\}_{n=1}^{N}$, where T_l^n indicates the lth local client currently receiving the task as the nth task and N indicates the total number of tasks. When C_l processes the current newly arrived task T_l^n (we set $n \geq 2$ for the convenience of describing the complete federated incremental task), C_l trains the local model $\theta_l^{n,r}$ (where $\theta_l^{n,r}$ represents the model of the lth local client, which processes task T_l^n in the rth global round) according to the following process.

1. C_l updates the example sample $\{X_{li}, Y_{li}\}_{i=1}^{m}$ in the dynamic example memory and preprocesses the new class sample $\{X_{li}^n, Y_{li}^n\}_{i=1}^{Nt}$ used by the current task T_l^n.

2. The example samples $\{X_{li}, Y_{li}\}_{i=1}^{m}$ together with the pre-processed new class samples $\{X_{li}^n, Y_{li}^n\}_{i=1}^{Nt}$ compose the local training dataset $\{X_l^n, Y_l^n\} = \{X_{li}, Y_{li}\}_{i=1}^{m} \cup \{X_{li}^n, Y_{li}^n\}_{i=1}^{Nt}$ for the task T_l^n.

3. The local dataset $\{X_l^n, Y_l^n\}$ is fed into the local model $\theta_l^{n,r}$, while at the same time $\{X_l^n, Y_l^n\}$ is fed into the best old model θ_l^{n-1}, and the corresponding classification result is obtained. This classification result is prepared for constructing the sample label smoothing loss L_{LS}.

4. The class gradient balance loss and sample label smoothing loss we constructed are used as the loss function of the local model $\theta_l^{n,r}$. $\theta_l^{n,r}$ is subsequently trained.

While $\theta_l^{n,r}$ is being trained, C_l selects a representative sample for each new class $K_l^{new} = \sum_{j=1}^{k} K_l^{n,j}$ (where K_l^{new} denotes all new classes received by the current task T_l^n, $K_l^{n,j}$ denotes the jth new class among all new

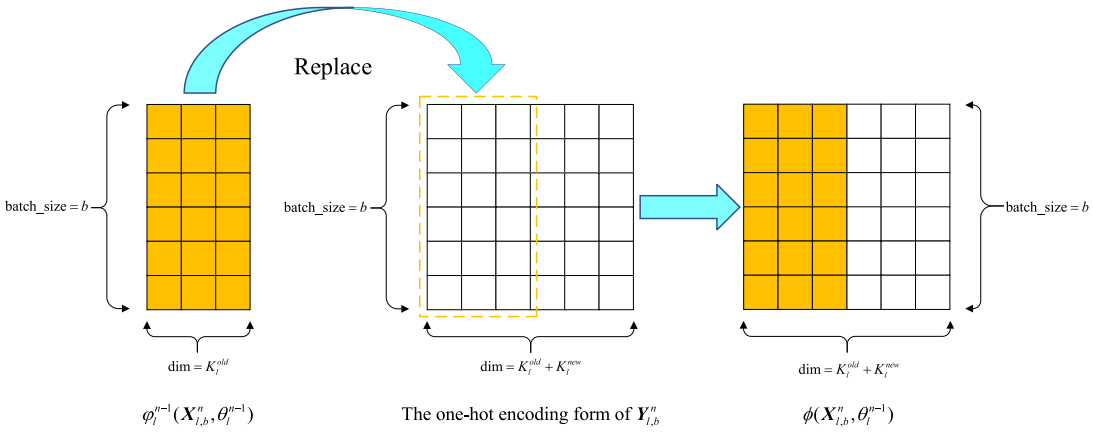


Fig. 2. Sample label smoothing operation.

classes, and k denotes the total number of new classes) in the current task T_l^n. Then, C_l sends the representative sample set $\{X_{l,R}^n, Y_{l,R}^n\} = \{x_{l,j}^n, y_{l,j}^n\}_{j=1}^k$ to the encoder model θ_l^E and collects the gradient information set. After the local model is trained, the local client uploads the model parameters to the global client C_g. At the same time, the local client also uploads the collected gradient information set to the relay client C_r. After completing the model aggregation and update, C_g distributes the global model $\theta^{n,r}$ to all local clients as well as the relay client C_r. Finally, C_r performs a sample reconstruction operation on all representative samples using the decoder θ_r^D with the same structure as in the local client, and evaluates θ_r^D using these samples as a test set. At the end of T_l^n, C_r selects the best-performing model θ_{best}^n among the global rounds occupied by T_l^n based on the evaluation metric of accuracy, and distributes θ_{best}^n to all local clients as the best old model when the next new task T_l^{n+1} arrives. With the above steps, we have completed the iteration and update of the FL-IIDS.

3.2. Incremental learning strategies for local clients

Three new mechanisms are designed in incremental learning strategies for local clients based on the analysis of the problems of catastrophic forgetting. They are named dynamic example memory, class gradient balancing loss, and sample label smoothing loss. The incremental strategy is designed to help local clients remember old classes, solving the problem of local models losing the ability to classify traffic data belonging to old classes in a constant stream of tasks.

3.2.1. Issue description

Issue 1: As described in Section 3.1, the local client in FL-IIDS needs to use the old samples in the example memory during training. The concept of example memory was born from the class incremental learning in the field of computer vision, named iCaRL [32]. Assuming the current task is T_l^n, the traditional example memory allocates storage space for each old class $K_l^{i,j}$ when creating an example sample set for the old classes $K_l^{old} = \sum_{i=1}^{n-1}\sum_{j=1}^{k} K_l^{i,j}$ (where $K_l^{i,j}$ represents the jth class in the ith task, k the total number of classes for each task, and $n-1$ represents the total number of old tasks) according to Eq. (1):

$$m_l^{i,j} = \frac{m}{(n-1)\times k} \tag{1}$$

where $m_l^{i,j}$ represents the storage space owned by the jth old class in the ith task, m represents the total size of the dynamic example memory. Subsequently, assuming that the data of the old class $K_l^{i,j}$ is represented as $x \in X$, according to the algorithm of iCaRL, the sample memory will select $m_l^{i,j}$ representative samples according to Eq. (2):

$$\begin{aligned} &\mu = \frac{1}{n}\sum_{i=1}^{n}\varphi_l(x) \\ &p_{k\in[1,m_l^{i,j}]} \leftarrow \underset{x\in X}{argmin}\left\|\mu - \frac{1}{k}\left[\varphi_l(x) + \sum_{j=1}^{k-1}\varphi_l(p_j)\right]\right\| \end{aligned} \tag{2}$$

where $\varphi_l(\cdot)$ represents the output of the local model (the local model referred to here is the model without the last softmax layer). However, this algorithm for creating example sample sets is not applicable to intrusion detection scenarios. Specifically, there is a class imbalance between various types of traffic data in a real-world intrusion detection application scenario. For example, normal traffic data and common attack traffic data make up the majority of the training set, while rare attack traffic data and unknown traffic data make up a smaller sample of the training set [33]. When the training samples $\{X_l^n, Y_l^n\} = \{X_{li}, Y_{li}\}_{i=1}^m \cup \{X_{li}^n, Y_{li}^n\}_{i=1}^{Nt}$ in C_l are involved in the local model training, the traditional example memory causes the local model to forget the real, unbalanced distribution among the old classes of samples in the intrusion detection system after training, which can lead to the catastrophic forgetting problem of the local client worsen.

Issue 2: In a real-world application environment of federated learning-based intrusion detection techniques, the local client is faced with a larger amount of data. The samples of old classes in the example memory are much less in number than the new classes samples for the current task, and for the deep learning model $\theta_l^{n,r}$ in C_l, there is a serious imbalance between the number of old and new classes' samples, and this serious imbalance will make the local model more inclined to learning new knowledge. Thus, the example memory construction cannot mitigate the local-side catastrophic forgetting problem of FL-IIDS very effectively.

Issue 3: Even though we have devised reasonable mechanisms to compensate for the catastrophic forgetting problem caused by this data imbalance for **Issue 2**, the multi-classification task is often more valuable to study in intrusion detection systems. The mechanism proposed for **Issue 2** lacks a specific means of memory for a specific old class. As

the task flow $T_l = \{T_l^n\}_{n=1}^N$ progresses, the sample size of each old class in the dynamic example memory will keep shrinking, which will lead to a gradual reduction of the learning ability of the local model for each old class. This phenomenon eventually dulls the local model's ability to recognize each specific old class, and may even lead to unexpected classification results.

3.2.2. Dynamic example memory

To solve Issue 1, a dynamic example memory is constructed, which makes the distribution of old class samples in the example memory more convergent to the true distribution. As receiving the task stream $T_l = \{T_l^n\}_{n=1}^N$, the local client records and saves the total number of all samples $\{Size[T_l^n]\}_{n=1}^N$ for each task and the number of samples $\{Size[K_l^{n,j}]\}_{j=1}^k$ for each class in the task, where k represents the total number of classes included in each task. When the local client processes the current task T_l^n, the example memory will allocate storage space for each old class according to Eq. (3):

$$m_l^{i,j} = \frac{mSize\left[K_l^{i,j}\right]}{\sum_{i=1}^{n-1} Size\left[T_l^i\right]} \tag{3}$$

where m represents the total size of the dynamic example memory. Since the data of each local client is distributed independently and identically among each other, whenever a local client receives a new class, the local client treats it as a signal that a new task has arrived. After the local client concludes that it is receiving a new task, C_l updates the example memory according to Eq. (3). As the task is continuously updated, the dynamic example memory ensures that the sample distribution of the old classes in the memory is the same as the sample distribution of the old classes that the local clients have already received for training. The dynamic example memory will prompt the local model to learn the correct sample distribution of traffic data and guide the local model to recall the real old class knowledge in the IDS, which further achieves the effect of strengthening the local model's memory of old classes and alleviating the catastrophic forgetting problem from the local client's perspective.

3.2.3. Class gradient balancing loss

To solve Issue 2, class gradient balance loss is constructed to constrain the learning rate for the new knowledge and the forgotten rate for the old knowledge.

Specifically, for a mini-batch of data $\{\boldsymbol{X}_{l,b}^n, \boldsymbol{Y}_{l,b}^n\} = \{x_{l,i}^n, y_{l,i}^n\}_{i=1}^b \subset \{X_l^n, Y_l^n\}$ in the task T_l^n being processed by the local client C_l (where b is batch-size and $y_{l,i}^n$ represents the true label of the ith sample), we first obtain the gradient information of the $y_{l,i}^n$th neuron $N_{y_{l,i}^n}^n$ in the last layer of the neural network of the local model $\theta_l^{n,r}$ via Eq. (4).

$$\begin{aligned} g_{l,i}^n &= \frac{\partial CE\left(\varphi_l^n\left(x_{l,i}^n, \theta_l^{n,r}\right), y_{l,i}^n\right)}{\partial N_{y_{l,i}^n}^n} \\ &= \varphi_l^n\left(x_{l,i}^n, \theta_l^{n,r}\right)_{y_{l,i}^n} - 1 \end{aligned} \tag{4}$$

where $CE(\cdot)$ is the cross-entropy loss function; $\varphi_l^n(\cdot)$ is the probability that the sample $x_{l,i}^n$ is obtained by $\theta_l^{n,r}$ prediction; $\varphi_l^n(x_{l,i}^n, \theta_{l,i}^n)_{y_{l,i}^n}$ represents the Softmax probability that sample $x_{l,i}^n$ is classified as into $y_{l,i}^n$ by $\theta_l^{n,r}$ prediction. To slow down the learning of K_l^{new} by the local model and enhance the memory of K_l^{old} by the local model, we calculate the mean value of the gradient information of the new classes and the old classes by Eqs. (5) and (6) and redefine the weights of the new category and the old category in the loss function by Eq. (7) to reweight the loss function.

$$g_{new} = \frac{\sum_{i=1}^b \left|g_{l,i}^n\right| \cdot \psi\left(y_{l,i}^n \in K_l^{new}\right)}{\sum_{i=1}^b \psi\left(y_{l,i}^n \in K_l^{new}\right)} \tag{5}$$

$$g_{old} = \frac{\sum_{i=1}^b \left|g_{l,i}^n\right| \cdot \psi\left(y_{l,i}^n \in K_l^{old}\right)}{\sum_{i=1}^b \psi\left(y_{l,i}^n \in K_l^{old}\right)} \tag{6}$$

$$L_{GB} = \frac{1}{b} \sum_{i=1}^b \frac{\left|g_{l,i}^n\right|}{g_{m,i}} \cdot CE\left(\varphi_l^n\left(x_{l,i}^n, \theta_l^{n,r}\right), y_{l,i}^n\right) \tag{7}$$

where $\psi(\cdot)$ is a conditional judgment function, when the variables in the function are judged to be true $\psi(\cdot) = 1$, otherwise $\psi(\cdot) = 0$; $g_{m,i} = \psi(y_{l,i}^n \in K_l^{new}) \cdot g_{new} + \psi(y_{l,i}^n \in K_l^{old}) \cdot g_{old}$. With the above approach, the sample imbalance between the new classes and the old classes is somewhat mitigated by the weights $\frac{\left|g_{l,i}^n\right|}{g_{m,i}}$. The weights are larger when the samples belong to the old classes which account for a smaller number of the current training samples, and vice versa. Based on the dynamic example memory, the class gradient balancing loss can actively balance the learning paces of new and old classes and further effectively mitigate the catastrophic forgetting of the local model.

3.2.4. Sample label smoothing loss

To solve Issue 3, we propose a knowledge distillation strategy called label smoothing loss, which main objective is to fuse old class knowledge in the form of soft labels into the labels of the current training data and to improve the model's ability to generalize to each old class through the distillation loss function, thus reducing the impact of catastrophic forgetting on the local model. Specifically, we first store the old model θ_l^{n-1} for the previous task T_l^{n-1} on the local client. Subsequently, we send the current mini-batch of data $\{\boldsymbol{X}_{l,b}^n, \boldsymbol{Y}_{l,b}^n\}$ to θ_l^{n-1} and the local model $\theta_l^{n,r}$, and obtain the output results $\varphi_l^{n-1}(\boldsymbol{X}_{l,b}^n, \theta_l^{n-1})$ for the old model θ_l^{n-1} and prediction results $\varphi_l^n(\boldsymbol{X}_{l,b}^n, \theta_l^{n,r})$ for the local model $\theta_l^{n,r}$. Next, we perform a smoothing operation on the label by replacing the first K_l^{old} dimensions of the hard label in the form of $\boldsymbol{Y}_{l,b}^n$'s one-hot encoding by $\varphi_l^{n-1}(\boldsymbol{X}_{l,b}^n, \theta_l^{n-1})$ as a soft label, and noting the newly formed label as $\phi(\boldsymbol{Y}_{l,b}^n, \theta_l^{n-1})$, as shown in Fig. 2. Finally, we perform knowledge distillation using KL divergence as shown in Eq. (8):

$$L_{LS} = KL\left(\varphi_l^n\left(\boldsymbol{X}_{l,b}^n, \theta_l^{n,r}\right) \middle\| \phi\left(\boldsymbol{Y}_{l,b}^n, \theta_l^{n-1}\right)\right) \tag{8}$$

With the sample label smoothing loss, the local model compensates for the deficiency that the class gradient balancing loss does not pay enough attention to the old classes of each kind and helps the local model to further solve the catastrophic forgetting problem for the old classes and achieve incremental learning on the local side.

In summary, we define the loss function used by the local model in the local client according to Eq. (9):

$$L_{C_l} = \gamma_{GB} L_{GB} + \gamma_{LS} L_{LS} \tag{9}$$

where γ_{GB} and are γ_{LS} hyperparameters. In the rth global round, we use

Eq. (9) as the optimization objective of the local model and train the local model, and the local client uploads the local model to the global client for global aggregation after the training. The training algorithm for the local model is shown in Algorithm 1

Algorithm 1: Incremental learning algorithms from a local client perspective

Input: training dataset $\{X_l^n, Y_l^n\}$, local clients $\{C_l\}_{l=1}^{L}$ participating in aggregation, all local clients $\{C_l\}_{l=1}^{K}$, current task T_l^n, best old model θ_{best}^n

Output: all local models $\{\theta_l^{n,r}\}_{l=1}^{L}$

1 :**if** detect the arrival of the current task T_l^n **then**
2 : **for** C_l in $\{C_l\}_{l=1}^{K}$ **do**
3 : Use Eq. (3) to dynamically adjust the number of samples for each old class;
4 : Select example samples $\{X_{li}, Y_{li}\}_{i=1}^{m}$ using Eq.(2);
5 : Store θ_{best}^n as the old model θ_l^{n-1};
6 : **end for**
7 :**for** C_l in $\{C_l\}_{l=1}^{L}$ **do**
8 : **for** $\{\boldsymbol{X}_{l,b}^n, \boldsymbol{Y}_{l,b}^n\}$ in $\{X_l^n, Y_l^n\}$ **do**
9 : Train local model $\theta_l^{n,r}$ via Eq. (9)
10: **end for**
11: **return** all local models $\{\theta_l^{n,r}\}_{l=1}^{L}$
11:**end for**
12:**end**

3.3. Incremental learning strategies from a global client perspective

Unlike the traditional incremental learning in the intrusion detection scenario, the incremental learning approach considering only the local side ignores the differentiation of local client types and the heterogeneity of data among local clients. It further leads to the problem that the global model generated after the global client completes global aggregation lacks the global perspective for solving catastrophic problems and is not strong in generalizing to old classes.

Specifically, we can classify all local clients in the FL-IIDS into three categories: C_o, C_m, and C_n [34]. C_o denotes a local client that does not receive the samples of new classes for the current task and only uses the samples of old classes from the dynamic example memory; C_m denotes a local client that receives the samples of new classes for the current task and also uses the samples of old classes from the dynamic example memory; and C_n denotes a new local client added while the current task is in progress, such a client only accepts the samples of new classes for the current task. The above three different classes of clients cause differentiation in the types of local clients. Although our study is based on the premise that the data are independently and identically distributed among local clients, we restrict each local client to see only a portion of the data for each new class. This condition leads to a slight difference between the old class samples and the new class samples among the local clients. The above situation results in the heterogeneity of data among the local clients.

3.3.1. Sample reconstruction based on relay client and model gradient

According to the description in Section 3.3, the construction of sample label smoothing loss requires C_l should store the old model θ_l^{n-1}. Intuitively, it is sufficient for each local client to find the best-performing model in the global round experienced by the current task and store it as the old model at the end of the current task. However, such an approach ignores the differentiation problem and heterogeneity we discussed above. Therefore, by finding the best old model from a global perspective, we can combine the information of all local clients involved in aggregation to obtain the old model with better generalization capability, thus solving the generation and spread of the global catastrophic forgetting problem.

Specifically, enlightened by [34], we construct a relay client C_r for finding the best old model from a global perspective and use a model gradient-based sample reconstruction algorithm to work with the relay client to discover the best old model. In the first global round when task T_l^n arrives, C_l picks the most representative sample for each of the new classes, where the representative sample is the one that is closest to the average potential feature space of all samples in that new class. Then, C_l generates a representative dataset $\{\boldsymbol{X}_{l,R}^n, \boldsymbol{Y}_{l,R}^n\} = \{x_{lj}^{'n}, y_{lj}^{'n}\}_{j=1}^{k}$ from the selected representative samples and their labels, and sends these data to the encoder model $\theta_l^E = \{W_i\}_{i=1}^{M}$ in C_l to calculate the gradient set $\{\nabla\theta_{lj}^E\}_{j=1}^{k}$. W_i represents the parameter of the ith layer in θ_l^E, and $\nabla\theta_{lj}^E$ denotes the gradient information of θ_l^E calculated by the jth class of representative data. For the ith element $\nabla_{W_i}\theta_{l,j}^E$ in the gradient information $\nabla\theta_{lj}^E$, we define it by Eq. (10):

$$\nabla_{W_i}\theta_{l,j}^E = \nabla_{W_i} CE\left(\varphi_l^E\left(x_{l,j}^{'n}, \theta_l^E\right), y_{l,j}^{'n}\right) \tag{10}$$

Then, C_l sends the gradient set $\{\nabla\theta_{lj}^E\}_{j=1}^{k}$ into C_r. This process ensures the privacy of the local side data in federal learning.

In C_r, we create the decoder network θ_r^D with the same encoder model as in the local client. C_r collects the gradient set $\nabla\theta^n = \bigcup_{l=1}^{L}\{\nabla\theta_{lj}^E\}_{j=1}^{k}$ sent by all local clients $\nabla\theta_{lj}^E$ participating in the global aggregation and infers the true value label y_t^n of this gradient information based on the last layer of gradient symbol information in $\nabla\theta_{lj}^E$ (proposed by [35]). Then, we generate initialized samples x_t^{*n} in C_r using the standard Gaussian distribution and send $\{x_t^{*n}, \nabla\theta_t^n, y_t^{'n}\}_{t=1}^{k\times L}$ to the decoder network (where t represents the tth element in $\nabla\theta^n$) and reconstruct each representative sample using Eq. (11):

$$\begin{aligned} L_{RS} &= \sum_{i=1}^{M}\left\|\nabla_{W_i} CE\left(\varphi_r^D\left(x_t^{*n}, \theta_r^D\right), y_t^{'n}\right) - \nabla_{W_i}\theta_t^n\right\|^2 \\ x_t^{*n} &\leftarrow x_t^{*n} - \gamma_{RS}\nabla_{x_t^{*n}} L_{RS} \end{aligned} \tag{11}$$

where γ_{RS} is the learning rate and $\varphi_r^D(\cdot)$ denotes the output result of θ_r^D. By the above method, we reconstruct a representative dataset for each local client in the relay client without leaking the local client data.

3.3.2. Selecting the best old model

After performing the sample reconstruction operation, the relay client uses the reconstructed sample dataset to evaluate the global model with accuracy for each global round and stores the best-performing model θ_{best}^n. When the local client detects the next new task T_l^{n+1}, C_r sends θ_{best}^n to all selected local clients in preparation for the construction of the sample label smoothing loss.

In summary, we have completed the whole design of the FL-IID architecture. The incremental learning algorithm from a global perspective is shown in Algorithm 2.

Algorithm 2: Incremental learning algorithm from a global perspective

Input: Gradient information set $\nabla\theta^n$

Output: the best model θ^n_{best}

1 :Create initialized sample $\{x_t^{*n}\}_{t=1}^{k\times L}$

2 :Extrapolation of ground truth labels $\{y_t^{'n}\}_{t=1}^{k\times L}$ based on gradient information

3 :**for** $\{x_t^{*n}, \nabla\theta_t^n, y_t^{'n}\}$ in $\{x_t^{*n}, \nabla\theta_t^n, y_t^{'n}\}_{t=1}^{k\times L}$ **do**

4 : Reconstruct representative data $\{\boldsymbol{X}_{l,R}^{'n}, \boldsymbol{Y}_{l,R}^{'n}\}$ via Eq. (11)

5 :**end for**

6 :Evaluate the model using the reconstructed data with accuracy as a criterion

7 :Store the best model θ^n_{best}

8 :**return** θ^n_{best}

9 :**end**

4. Experiment and discussion

In this section, we elaborate on the details of the dataset required for the experiments, the experimental setup, and the evaluation metrics. Based on this, we will analyze in detail the results of comparative experiments, ablation experiments, and other experiments to verify the validity and rationality of our proposed mechanism.

4.1. Selection and division of datasets

We verify our proposed mechanism using the UNSW-NB15 dataset and the CICIDS2018 dataset. The UNSW-NB15 dataset and the CICIDS2018 dataset are commonly used for evaluating FL-based intrusion detection systems. UNSW-NB15 is an intrusion detection dataset collected and collated by Moustafa [36] et al. at the Australian Cyber Security Centre. The UNSW-NB15 dataset covers a more comprehensive range of modern cyberattacks. Besides of the normal traffic, it contains nine classes of the attack traffic. The original UNSW-NB15 dataset contains a large number of normal instances, and this imbalance between normal and attack instances tends to lead the model to fit normal instances more [1]. Therefore, we choose the UNSW-NB15-training dataset and the UNSW-NB15-testing dataset. These two datasets are subsets of the original UNSW-NB15, which are extracted from the original UNSW-NB15 dataset, retaining most of the attack examples while removing some normal examples and ensuring that the training and testing sets have similar distributions [37].

The CICIDS2018 dataset is the latest dataset developed by the Canadian Institute of Cyber Security (CIC) [38]. The traffic characteristics of this dataset fit modern network traffic and can effectively imitate realistic traffic so as to evaluate our work. The dataset includes six attack classes and one normal class.

We use the UNSW-NB15 dataset to simulate the situation where the number of samples varies significantly between different attack categories. The dataset contains 46 features, where the features ‘proto’ ‘service’ ‘state’ are features belonging to the character category. As numeration of character features produces more redundant features. And our work does not use any feature reduction method. Since we remove these three features to avoid the effect of feature redundancy on the model. To avoid the class imbalance problem from interfering with the experiments, we remove the class ‘worm’ from the dataset and divide the remaining nine classes into three tasks, each containing three classes. At the same time, we put the ‘normal’ class in the first task phase to simulate the reality where the normal class of traffic data appears first and most often. The subsequent task phases have only the attack classes to simulate the real application environment where the model will be constantly exposed to new attacks. See Table 1 for the specific partitioning results of the UNSW-NB15 dataset.

Table 1
Division details of the UNSW-NB15 dataset.

Task Phase	Traffic type	Size of the training set	Size of the testing set
Task 1	Normal	56000	36988
	Analysis	2000	677
	Backdoor	1746	583
Task 2	Dos	12264	4089
	Exploits	33393	11132
	Fuzzers	18184	6062
Task 3	Generic	40000	18871
	Reconnaissance	10491	3496
	Shellcode	1133	378

Table 2
Division details of the CICIDS2018 dataset.

Task Phase	Traffic type	Original dataset size	Adjusted dataset size
Task 1	Benign	6112151	61122
	DDoS	687742	6877
	Infiltration	161934	1619
Task 2	Bot	286191	2862
	Brute-Force	380949	3809
	DoS	654300	6543

We use the CICIDS2018 dataset to simulate the situation where the sample sizes are relatively close between different attack classes. The dataset contains 79 features. Since there are no character features among the features, we do not perform any censoring on the features in this dataset. To facilitate the analysis of experimental results with the UNSW-NB15 dataset, we reduce the size of each category in the CICIDS2018 dataset and adjust the sample size to an order of magnitude similar to that of the UNSW-NB15 dataset. This was done by randomly selecting 1% of the data for each class sample in CICIDS2018. Subsequently, the training set and test set were divided according to a 4:1 ratio. Further, we discard the Web category attacks in the CICIDS2018 dataset and divide the remaining six classes into two tasks, each containing three classes. Similarly, we have placed normal traffic in the first task phase. See Table 2 for the specific partitioning results of the CICIDS2018 dataset.

4.2. Experimental environment and parameter settings

Our experimental environment is summarized as follows. The processor is AMD Ryzen 5800H with 16 GB of RAM and the graphics card is NVIDIA GeForce RTX 3070 with 8 GB of RAM. The development language is Python 3.7.13 for Windows 11 and the neural network framework is Pytorch, version 1.10.0. Our federated learning framework is built entirely on Pytorch and does not use any third-party libraries

The FedAvg [39] is chosen as the federated learning framework and conducts a series of experiments on this federated learning framework using the two datasets introduced in Section 4.1. The workflow in the FedAvg framework is as follows:

(1) The global model is initialized
(2) The global client sends the model to the local client with global model parameters and selects some of the local clients to participate in the training

Table 3
Meaning of each parameter in the evaluation metric.

Parameters	Specific meaning
T_P	Positive instances are judged to be positive instances
T_N	Positive instances are judged as negative instances
F_P	Negative instances are judged as positive instances
F_N	Negative instances are judged as negative instances

(3) The selected local clients update the local model with the received global model parameters and train with local data
(4) Selected local clients upload local model parameters to the global client
(5) The global client updates the global model by weighted averaging. Subsequently, steps 2 through 5 are repeated

For the UNSW-NB15 dataset, we initialize 10 local clients, set the global rounds to 15, and train 5 global rounds for each task. In each global round, we randomly select 3 local clients to participate in model aggregation. In the local clients, the number of training rounds for the local model in each global round is 20, and we use an SGD optimizer with a learning rate of 1.0 to train our local model. The size of the dynamic example memory is set to 2% of the size of the UNSW-NB15-training dataset. Encoder and decoder are used to collect gradient information for sample reconstruction using a simple fully connected layer-activation function layer-fully connected layer structure. In the process of sample reconstruction for relay clients, we use an L-BFGS optimizer with a learning rate of 0.1 to generate reconstruction samples, and the number of iterations is set to 250. To achieve the differentiation of local client types, when a new task comes, we add a new local client as C_n among all the original local clients, and we randomly select a client as C_o among the remaining local clients. To facilitate the analysis and comparison of the experimental results, for the CICIDS2018, we set the number of global rounds to 10, trained 5 global rounds for each task, and the local client was trained using an SGD optimizer with a learning rate of 0.1. The rest of the experimental parameters were set the same as UNSW-NB15.

Since the direction of our study and analysis is the FL-IIDS in the case where the data distribution among the local clients is independently and identically distributed, for the two datasets UNSW-NB15 and CICIDS2018, we set the local clients participating in the model aggregation to receive all traffic categories in the current task, but each local client participating in the aggregation can only randomly receive thirty percent of the data of each class, this practice is to achieve heterogeneity of data among local clients. Also, to facilitate the analysis of the results of the ablation experiment, we set both hyperparameters γ_{GB} and γ_{LS} of Eq. (9) to 0.5.

4.3. Evaluation metrics

We mainly evaluate our proposed FL-IIDS using the following three metrics. They are accuracy, precision, and recall, respectively. Specifically, accuracy is used to evaluate the overall classification effectiveness of the global model on the test set for all classes of samples, while precision and recall are used to evaluate the classification performance of the global model for each class of traffic samples in the test set. These evaluation metrics are calculated as follows:

$$\text{Accuracy} = \frac{T_p + T_N}{T_p + T_N + F_P + F_N} \tag{12}$$

$$\text{Precision} = \frac{T_p}{T_p + F_p} \tag{13}$$

$$\text{Recall} = \frac{T_P}{T_P + F_N} \tag{14}$$

Eqs. (12), (13), (14) in the meaning of the specific parameters are shown in Table 3.

4.4. Comparison experiment

4.4.1. Comparison experiments of classification models

We verify the effectiveness and generalization of our proposed FL-IIDS by comparing the performance of different classification models used for training. Table 4 shows the performance of our proposed mechanism on the UNSW-NB15 and the CICIDS2018, and the models involved in the comparison are DNN [40], RNN [41], CNN (two layers) [42], and CNN-GRU [43]. Analyzing the experimental results, we can draw the following conclusions. First, for UNSW-NB15 and CICIDS2018, our proposed FL-IIDS achieves better performance on the first task. This experimental result shows that neither the change of loss function due to the local incremental learning strategy nor the addition of relay servers due to the global incremental learning strategy affects the classification accuracy of the global model for the first task. This finding also shows from the side that our proposed mechanism is effective for the federal intrusion detection task under non-incremental learning conditions. Second, for UNSW-NB15, our proposed mechanism can largely alleviate the catastrophic forgetting problem caused by incremental learning patterns in federal IDS. Specifically, the accuracy of the four classification models in task two decreases from 29.127% to 33.197% compared

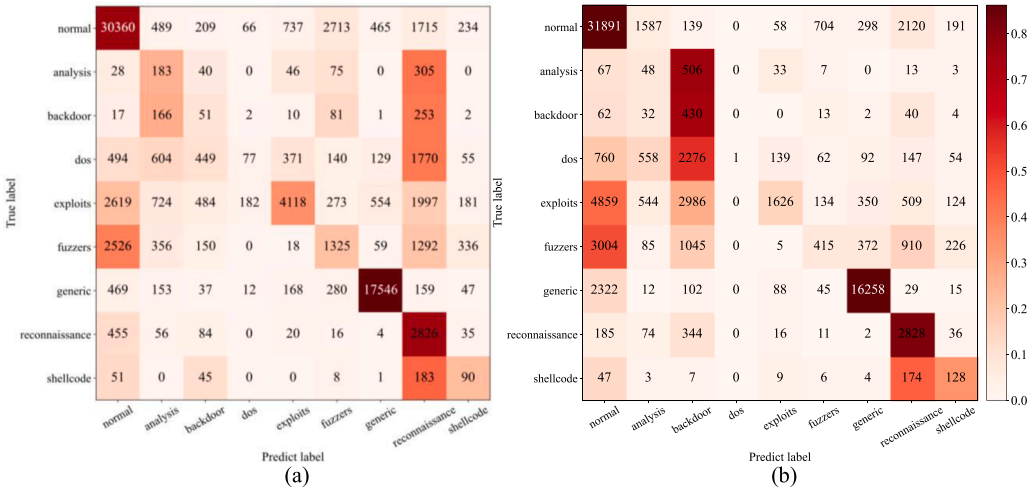


Fig. 3. Confusion matrix comparison results using different example memories on the UNSW-NB15.

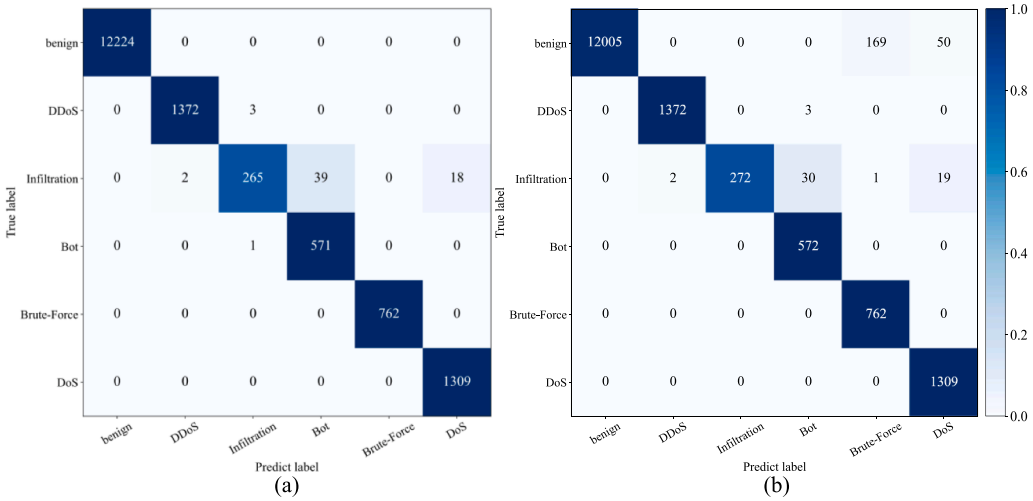


Fig. 4. Confusion matrix comparison results using different example memories on CICIDS2018.

Table 4
Comparison of experimental results of classification models for incremental intrusion detection mechanism based on federal learning.

Dataset	Classification Model	Accuracy at different task phases(%)		
		Task 1	Task 2	Task 3
UNSW-NB15	DNN	96.504	63.307	62.917
	RNN	93.281	61.464	55.689
	CNN	95.145	64.377	62.050
	CNN-GRU	96.583	69.456	68.764
CICIDS2018	DNN	99.993	99.565	
	RNN	98.197	97.217	
	CNN	100	99.626	
	CNN-GRU	100	99.620	

to task one. The accuracy in Task 3 decreased from 0.390% to 5.775% compared to Task 2. Further analysis shows that the larger decrease in accuracy in task 2 is because the three old classes contain a large number of samples of the Normal class during the training process. With the combined effect of relay client and sample label smoothing loss, some samples of the DoS class, Exploits class, and Fuzzers class in Task 2 were classified as Normal class and thus resulted in a larger decrease in accuracy. In Task 3, as three more categories are added to the old category, the best old model selected by the relay client is no longer biased towards the Normal class, which further leads to the sample label smoothing loss without causing a large number of samples from the three new categories in Task 3 to be classified into the wrong old class. Third, for CICIDS2018, our proposed mechanism performs extremely well under the evaluation metric of accuracy, with only a performance degradation ranging from 0.374% to 0.980% for Task 2 relative to Task 1. On the one hand, the training set of this dataset has the same distribution as the test set, and no concept drift problem occurs. On the other hand, a large number of 0 features appear in each traffic data due to the unique thermal coding of UNSW-NB15 after data preprocessing. There is no feature extraction link among our proposed mechanism, which will lead to a large number of redundant features interfering with the learning of the model. In CICIDS2018, on the other hand, since there is no character features in the data, there is no need for a solo-heat coding session, and there are relatively few redundant features. All these factors contribute to the excellent performance of our proposed mechanism at CICIDS2018. Fourth, according to the experimental results of different classification models in Table 4, it is demonstrated that our proposed mechanism can be applied to a variety of classification models with certain generalization abilities under the FedAvg framework.

Table 5
Final accuracy comparison results of different example memories on two data sets.

Dataset	Methods	Final Accuracy (%)
UNSW-NB15	Ours	68.764
	Original	65.177
CICIDS2018	Ours	99.620
	Original	98.346

4.4.2. *Example memory comparison experiment*

To test the advantages of our proposed dynamic example memory over the original example memory, we conducted example memory comparison experiments. Fig. 3 shows the comparison results of the confusion matrix of the two example memories on UNSW-NB15. Among them, Fig. 3(a) shows the results using our proposed FL-IIDS, and Fig. 3 (b) shows the results of changing the dynamic example memory to the original example memory. Fig. 4 shows the comparison results of the confusion matrix of these two example memories at CICIDS2018. In particular, Fig. 4(a) shows the results for FL-IIDS and Fig. 4(b) shows the results for changing the dynamic example memory to the original example memory. Table 5 shows the final accuracy comparison results for different memories on both datasets. The classification models used for the above comparison results are CNN-GRU.

Through Table 5 we can conclude that the dynamic example memory can improve the classification accuracy of the local model to a certain

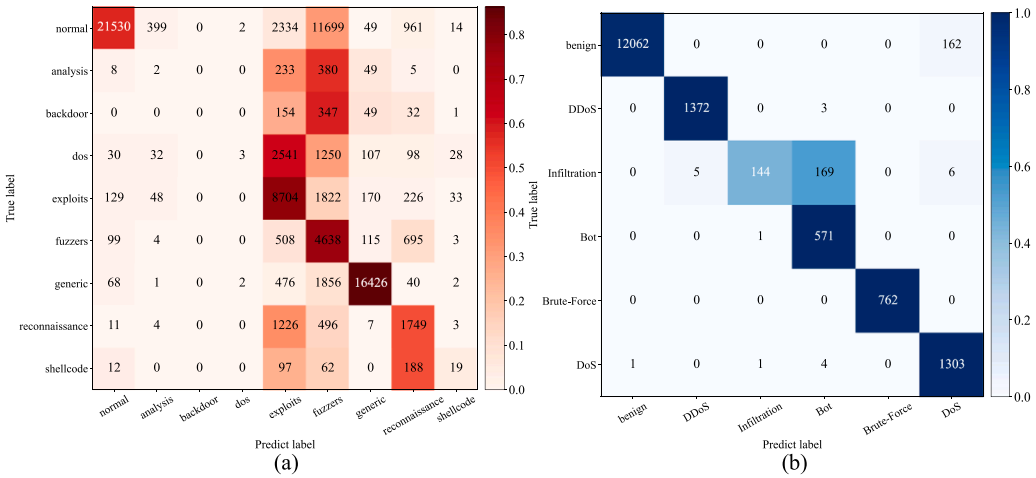


Fig. 5. Results of ablation experiments on class gradient balance loss.

extent, and, improve the classification accuracy of the global model under the federal framework after the local model is continuously aggregated. Further, we observe the results of the confusion matrix comparison between Figs. 3 and 4, and we can find that the construction of dynamic example memory can effectively help the federal learning framework recall the correct and true distribution of old class data and reduce the risk of forgetting of the global model. For example, as shown in the results in Fig. 3(b), because the original example memory has the same number of samples for each old class, the local client does not learn the correct old class sample distribution, and after several model aggregation updates results in the number of Exploits and Fuzzers classes, which are old classes with relatively large training samples, eventually being correctly identified by the global model relative to the dynamic example memory by 60.515% and 68.679%.

4.5. Ablation experiments

In this subsection, we will conduct exhaustive ablation experiments on the proposed mechanism. To ensure the fairness of the ablation experiments, all experiments in this summary use CNN-GRU as the classification model.

4.5.1. The role of class gradient balance loss

To explore the role of the class gradient balancing loss on our proposed FL-IIDS, in this subsection, we remove the class gradient balancing loss in Eq. (9) used in the local client training model and replace it with the ordinary cross-entropy loss, with the coefficient before the ordinary cross-entropy loss still set to 0.5, and evaluate the performance on both datasets. Fig. 5 shows the confusion matrix results of this experiment on both datasets. Among them, Fig. 5(a) shows the confusion matrix results obtained on the UNSW-NB15 dataset without using the class gradient balance loss and replacing it with the ordinary cross-entropy loss, and Fig. 5(b) shows the results obtained on the CICIDS2018 dataset.

We compare and analyze Fig. 5 with Fig. 3(a) (confusion matrix result of FL-IIDS when using UNSW-NB15) and Fig. 4(a) (confusion matrix result of FL-IIDS when using CICIDS2018) obtained using the full mechanism, and we can obtain the following two conclusions. First, the class gradient balance loss can help the local model to better remember old knowledge during training under the condition that the sample of old class examples is small, alleviating the catastrophic forgetting on the local side, which in turn improves the global model's memory of old knowledge after the local model is continuously aggregated and updated. On UNSW-NB15, the classification accuracy of the global model for all old classes is 60.664%, and that of the global model with the class gradient balance loss removed and replaced by ordinary cross-entropy is

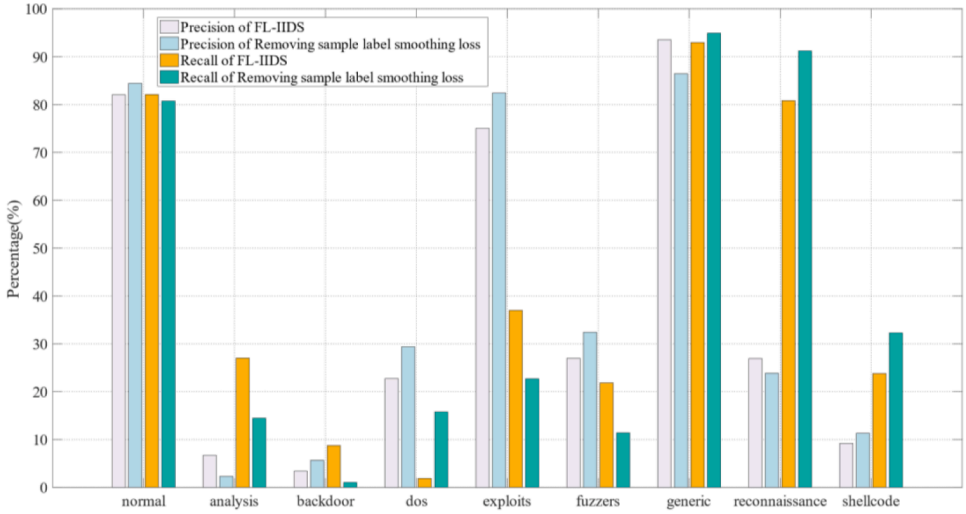


Fig. 6. Results of ablation experiments for sample label smoothing loss on UNSW-NB15.

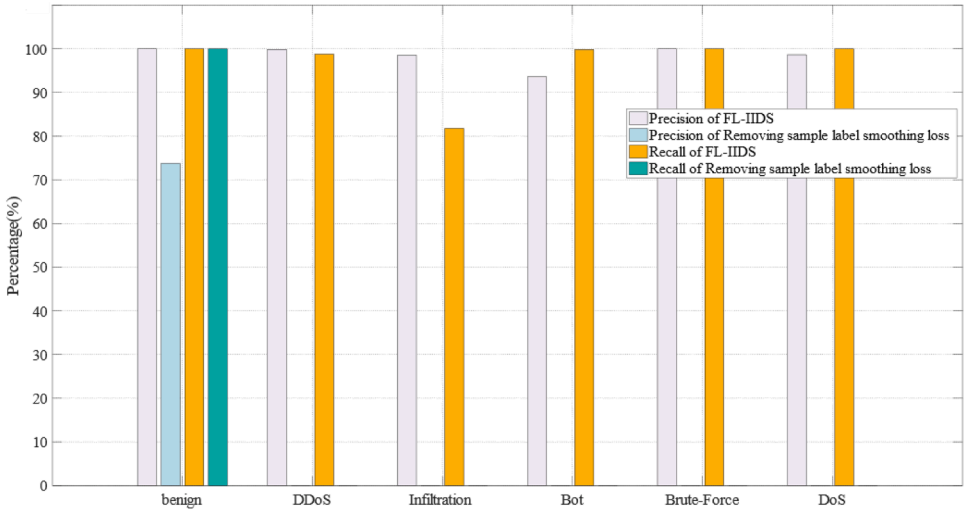


Fig. 7. Results of ablation experiments for sample label smoothing loss on CICIDS2018.

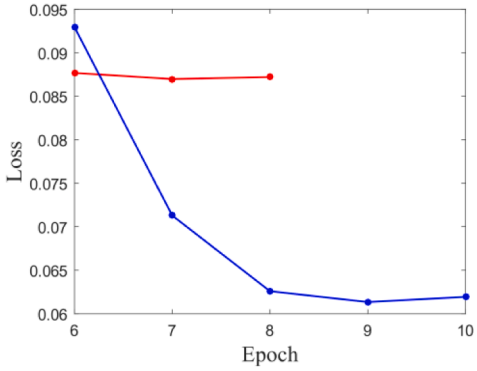


Fig. 8. Results of ablation experiments for sample label smoothing loss on UNSW-NB15.

58.586% for all old classes. And the latter's ability to classify the three old classes belonging to Task 1 was significantly reduced. It can be concluded that the absence of class gradient balance loss leads to a reduction in the overall old category recognition ability of the incremental intrusion detection mechanism based on federal learning, and the global model presents a more severe forgetting of data from older classes with a more distant period as the task is continuously updated. At CICIDS2018, the classification accuracy of the global model for all old classes was 99.555%, and the global model with the class gradient balance loss removed and replaced by ordinary cross entropy was 97.522% for all old classes. Second, by observing the confusion matrix, we can learn that the category gradient balance loss does not affect the learning ability of the local model for the data of the new classes of the current task, further improving the classification ability of the global model for the data of the new classes of the current task.

4.5.2. The role of sample label smoothing loss

In this subsection, we remove the sample label smoothing loss from Eq. (9) used in the local client training model and compare the results of this ablation experiment on two datasets. Fig. 6 shows the precision and recall for each class on the UNSW-NB15. Fig. 7 shows the results for the CICIDS2018.

By analyzing the experimental results, we can get the following two conclusions. First, according to the recall comparison results of the global model on UNSW-NB15, the sample label smoothing loss can help the federal learning framework to improve the recall of traffic data from old classes, but the precision comparison results also indicate that the sample label smoothing loss reduces the ability of the framework to accurately identify traffic data from old classes and may lead to a higher false alarm rate. At the same time, because the sample label smoothing loss changes the hard labels of the training samples in the local client to soft labels with the old class label information, the global model ends up with a slightly worse recall for the current new class data than without the sample label smoothing loss. Second, at CICIDS2018, after removing the sample label smoothing loss, the final recall and accuracy results of the global model show a more extreme situation, and it is easy to see that the reason for this situation is that all three old classes and three new classes are classified into Benign classes. Further analyzing the reasons for this situation, firstly, the absence of sample label smoothing loss would prevent the samples involved in training in the local client from fusing specific old class knowledge, further leading to the local model incorrectly classifying all the old category data as the Benign class with a high percentage of dynamic example memory during training. Secondly, the training data of Task 2 in the CICIDS2018 is much less in quantity than that of Task 1, mainly because there is more normal class traffic in the intrusion detection task. Comparatively, samples from the older classes in the dynamic example memory account for a larger proportion of the training data relative to the UNSW-NB15. This situation may eventually lead to poorer data classification accuracy of the global model for the new classes of the current task.

To further explore the reasons for the extremes in the CICIDS2018 dataset and the impact of the label smoothing loss on the convergence speed of the local model. We plot the final loss averages of local clients participating in training from the 6th global epoch to the 10th global epoch as shown in Fig. 8, where the blue folds represent FL-IIDS and the red folds represent the sample label smoothing loss removed.

As shown in Fig. 8, the local model of FL-IIDS starts to converge at the beginning of the eighth global epoch on the CICIDS2018 dataset. And after removing the sample label smoothing loss, the local model converged at the beginning of the sixth global epoch. It can be seen that although our method can help the model memorize the old classes, it will affect the convergence speed of the local model to some extent. However, although our method converges slightly slower, the red line

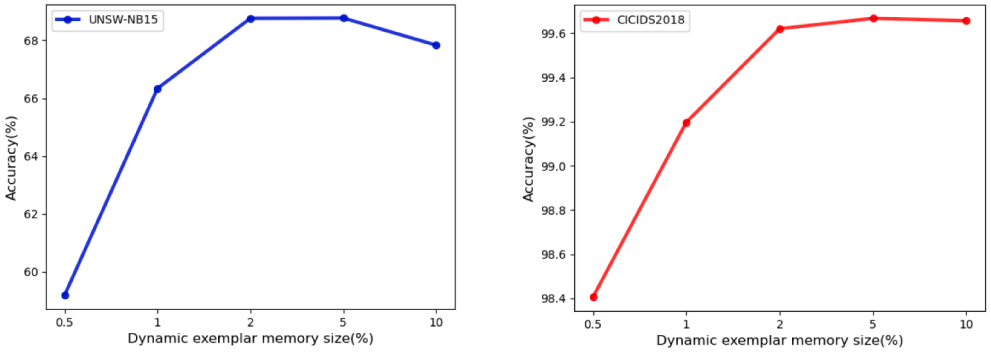


Fig. 9. Experimental results of capacity comparison of dynamic example memory.

Table 6
Results of ablation experiments with global incremental strategy.

Dataset	Methods	Evaluation Indicators		
		Precision (%)	Recall (%)	Accuracy (%)
UNSW-NB15	Ours	38.496	41.802	68.764
	Without *Cr*	38.213	37.660	68.748
CICIDS2018	Ours	98.436	96.900	99.620
	Without *Cr*	98.117	96.351	99.505

shows that the loss value of the local model with the sample label smoothing loss removed becomes "NaN" at the 9th global epoch. This is also in line with the more extreme experimental results in Fig. 7, which illustrates the necessity of the sample label smoothing loss.

4.5.3. Ablation experiment for global incremental strategy

In this subsection, we analyze and discuss the final performance of the global model under the condition that the relay client is not used to pick the best old model for each local client. Table 6 shows the final classification performance of the global model on the two datasets. In this case, both precision and recall are the average results of precision and recall for all classes in the dataset.

According to the results in Table 6, the average precision, the average recall, and the final accuracy of the global model show a small decrease when the best old model is selected for each local client without using the relay client. And on both datasets, the average recall rate decreases more compared to the precision rate. The main reasons for these phenomena are as follows. First, since our research is oriented towards incremental intrusion detection based on federated learning with independent and identically distribution of data among local clients. Although we simulate the data heterogeneity phenomenon and client differentiation phenomenon by some methods, the relay client can stand in the global view to pick the best old model for all local clients, which can alleviate the proliferation of the global catastrophic forgetting phenomenon to some extent. However, the best old model selected by the relay client is in essence little different from the best old model that each local client only stores relative to itself. The incremental strategy from the global perspective has a limited effect on the improvement of the overall federal learning framework. Second, the best old model selected by the relay client has a stronger generalization capability, which is passed on to the local client among the sample label smoothing losses and further improves the detection capability of the local model for each true positive sample. Eventually, after the continuous aggregation and updating of the federal framework, the global model presents a recall rate, and the precision rate is slightly higher than the case without the relay client.

4.6. Experiments on the capacity of the dynamic example memory

Fig. 9 shows how the accuracy metrics corresponding to our proposed FL-IIDS change as the dynamic example memory size is changed. During the experiments, we choose CNN-GRU as the classification model. Fixing other experimental parameters constant, the size of dynamic example memory is changed to 0.5%, 1%, 2%, 5%, and 10% of the training set. Among them, Fig. 3(a) shows the variation process of the final accuracy of the global model on the UNSW-NB15 test set, and Fig. 3 (b) shows the variation process of the final accuracy of the global model on the CICIDS2018 test set. By analyzing the experimental results, we can conclude that until the size of dynamic example memory is 5% of the training set size, the global model keeps getting better at classifying traffic data as the memory size of the dynamic example memory keeps getting larger. After the size of dynamic example memory exceeds 5%, the classification accuracy of the global model decreases to some extent. The reason for this phenomenon is that the larger memory size helps the global model to recall the old knowledge better, however, once a certain threshold is exceeded, the model fits the old knowledge too well and further causes the new category samples of the current task to be judged as the old category in the test and thus the accuracy decreases.

5. Conclusion

This work builds an intrusion detection system called FL-IIDS, which remembers knowledge about old classes in the data stream and solves the problem of catastrophic forgetting. We first improve the traditional example memory to make the data distribution in the memory more realistic. Second, we combine the class gradient balance loss and the sample label smoothing loss, which together constitute the loss function of the local model to solve the catastrophic forgetting in the local perspective. Finally, we construct the relay server and use the sample reconstruction method to solve the spread of catastrophic forgetting from the global level. Through experiments, we demonstrate that FL-IIDS can solve the catastrophic forgetting problem faced by federated learning-based IDS. In the future, we will work on federated incremental intrusion detection systems in the case of non-independent and identically distributed (non-i.i.d.) data across local clients.

CRediT authorship contribution statement

Zhigang Jin: Conceptualization, Methodology, Funding acquisition, Writing – review & editing. **Junyi Zhou:** Methodology, Software, Formal analysis, Writing – original draft. **Bing Li:** Writing – review & editing, Supervision. **Xiaodong Wu:** Writing – review & editing, Investigation. **Chenxu Duan:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets we use are all public authoritative datasets.

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